

Section-B

Note: Short answer question. Attempt any six questions out of eight questions. (6x5=30)

Q.11 If $A = \begin{bmatrix} 2 & 1 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -3 \\ 2 & 5 \end{bmatrix}$ then find $A - 2B$

Q.12 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Q.13 Differentiate $y = x \sin x$ with respect to x

Q.14 Find the value of x and y so that the vectors $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ are equal.

Q.15 Evaluate $\int (2x^2 + 5 \cos x + a^x + 1) dx$

Q.16 Evaluate $\int_0^1 (2x^2 + 3x) dx$

Q.17 Find the equation of straight line passing through $(1, 1)$ and making angle 30° with x axis.

Q.18 Is the function $f(x) = x^2$ one-one? Justify your answer.

Section-C

Note: Long answer questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Find the all points of maxima and minima and their corresponding maximum and minimum value of the function $y = -x^3 + 12x^2 - 5$

Q.20 Solve the following Linear Programming Problems graphically:

Maximise $Z = 5x + 10y$ subject to the constraints:
 $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y > 0$, $x, y \geq 0$.

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**Level 3, 2nd Semester / DVOC / (Ref. & Air Cond.,
Auto Servicing, ITM, PT, SD, AMT, FP, EMS, GM**

Subject : Applied Mathematics-II

Time : 2 Hrs.

M.M. : 50

Section-A

Note: Very short answer questions. Attempt all ten questions. (10x1=10)

Q.1 Write an example of a diagonal matrix.

Q.2 Evaluate $\begin{vmatrix} 8 & -2 \\ 1 & 3 \end{vmatrix}$

Q.3 If $f(x) = 2x^2 + 4x + 1$ find $f(1)$

Q.4 The value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is

Q.5 $\frac{d}{dx}(\sin x) =$ _____

Q.6 $\int x^2 dx =$

Q.7 $\int_0^{\pi} \sin x dx =$ _____

Q.8 The order of the differential equation:
 $\frac{d^2 y}{dx^2} + y = 2 \sin x$ is _____

Q.9 Write the slope of the straight line making angle of 30° with x -axis.

Q.10 Define Equal Vectors.